

Copying/syncing files and directories locally

Copying a file

The following command will sync a single file on a local machine from one location to another. Here the source file `test_bk.zip` is being synced to `/tmp/backups`.

```
$: rsync -zvh test_bk.zip /tmp/backups/
```

In the above command, `-zvh` are three different options where 'z' compresses file data, 'v' stands for verbose and 'h' outputs numbers in a human-readable format. After running the command, you will get an output similar to what follows:

```
$: rsync -zvh test_bk.zip /tmp/backups/
test_bk.zip
```

```
sent 2.82M bytes  received 35 bytes  5.64M bytes/sec
total size is 3.73M  speedup is 1.32
```

Copying/syncing a directory

The following command will transfer or sync all the files from one directory to a different. Here, we are copying the directory `/test` to `/tmp.backup`.

```
$: rsync -avzh test /tmp/backups/
```

Result:

```
$: rsync -avzh test /tmp/backups/
sending incremental file list
test/
test/dropbox_test.py
test/flask_app.py
test/flask_app_bk.py
```

```
sent 6.38K bytes  received 77 bytes  12.91K bytes/sec
total size is 25.05K  speedup is 3.88
```

Copying/syncing files and directories to or from a remote server

Copying a directory to a remote machine

The following command will sync a directory from a local to a remote machine. The command now copies the directory `/test` from our local machine to the home directory on the remote machine.

```
$: rsync -avz test/ root@xxx.xxx.xxx.xxx:/home/
```

Here `xxx.xxx.xxx.xxx` is the IP address of the remote machine.

Result:

```
$: rsync -avz test/ root@xxx.xxx.xxx.xxx:/home/
root@xxx.xxx.xxx.xxx's password:
sending incremental file list
```

```
./
dropbox_test.py
flask_app.py
flask_app_bk.py
```

```
sent 6,364 bytes  received 76 bytes  858.67 bytes/sec
total size is 25,049  speedup is 3.89
```

Copying a remote directory to a local machine

To copy a directory from a remote server to a local machine, use the following command:

```
$: rsync -avzh root@xxx.xxx.xxx.xxx:/home/test /tmp/backup_
remote
```

Result:

```
$: rsync -avzh root@xxx.xxx.xxx.xxx:/home/test /tmp/backup_
remote
root@xxx.xxx.xxx.xxx's password:
receiving incremental file list
test/
test/test.css
test/test.html
test/test.js
```

```
sent 3.79K bytes  received 10.71k bytes  228.42K bytes/sec
total size is 76.99M  speedup is 44.94
```

Using Rsync and SSH together

SSH, which stands for Secure Shell, is a cryptographic network protocol for initiating text-based shell sessions on remote machines in a secure way. It was designed and created to provide the best security when accessing another computer remotely. SSH provides security by encrypting the sessions and hence when you use Rsync over SSH, all of your data is encrypted and cannot be easily read by others.

Rsync allows us to copy files recursively with compression and over an encrypted channel. And to do so, we use Rsync with the `-e` option. The following command copies the file `test_bk.zip` from the local machine to the remote server over SSH:

```
$: rsync -vzhe ssh test_bk.zip root@xxx.xxx.xxx.xxx:/home/
```

Result:

```
$: rsync -vzhe ssh test_bk.zip root@xxx.xxx.xxx.xxx:/home/
root@xxx.xxx.xxx.xxx's password:
```

```
sent 45 bytes  received 12 bytes  8.77 bytes/sec
total size is 3.73M  speedup is 65,426.14
```

Similarly, files and directories can be copied from remote